Wafer nuisance filter: how many labels does it need?

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1 Introduction

Die-to-die inspection compares each die with a reference die and flags where they differ. Most differences are *nuisance* (misalignment residue, linewidth variation, illumination), so a second stage, the nuisance filter, decides which candidates are real defects. It is trained on reviewed, labeled candidates, and labels are expensive: how many does it need?

This write-up builds such a pipeline on synthetic dies and compares what standard learning theory says with what is measured, in four results: a VC bound for a threshold on peak contrast (Section 3); VC and Rademacher bounds for a linear filter (Section 4); the asymptotic learning curve of logistic regression, and its use to predict what more labels will buy (Section 5); and a certificate for the die-level false-alarm rate (Section 6). Section 7 collects where the bounds lose. Every number below is generated from a results file of the code.

2 Setup

Dies and stage 1. A die is a 256×256 grayscale image of a product layout with per-die nuisance (a smooth local warp that rigid alignment cannot remove, linewidth variation, intensity blobs, line-edge roughness) and, on half of the dies, one defect (particle, scratch, bridge, open or missing contact). Stage 1 aligns the die to a reference (the mean of 32 defect-free dies) by FFT cross-correlation with sub-pixel refinement and a least-squares gain/offset match, smooths $|\text{test} - \text{reference}|$ with a Gaussian ($\sigma = 1\,\text{px}$), thresholds it at $\tau = 0.0597$ and takes 8-connected components as candidates. τ was tuned on 4,000 dies for recall 0.990 (out of sample: 0.9909). A candidate is a real defect ($y = +1$) iff it overlaps the defect mask.

Stage 2. A threshold on peak contrast (Result 1), or a linear classifier $x \mapsto \text{sign}(w \cdot x)$ on k features plus an intercept (Results 2–4). The features are 10 hand-crafted ones (area, contrasts, shape, gradient energy, Laplacian of Gaussian at two scales, distance to the nearest pattern edge) or a frozen ResNet-18 embedding of the difference over the candidate ($d = 384$), optionally reduced by PCA to 4, 16 or 64 dimensions. All are standardized, clipped and rescaled so $\|x\| \le C = 1$. A die is flagged at threshold t iff some candidate's score exceeds t.

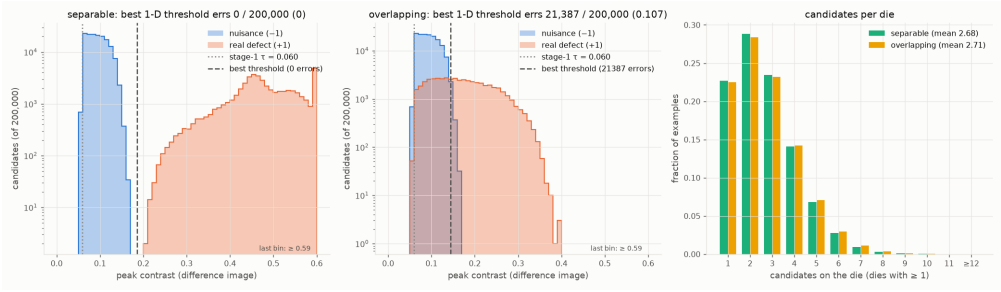


Figure 1: Peak contrast of nuisance and real-defect candidates in the two regimes (population pool) and candidates per die.

Sampling. Candidates on one die are dependent, so each example is one candidate drawn uniformly from a freshly generated die (dies without candidates are redrawn). With D a random die and $C(D)$ its candidates, examples are i.i.d. from

$$Q(A) = \frac{\mathbb{E}\big[\,|C(D) \cap A| \,/\, |C(D)| \;;\; |C(D)| \geq 1\big]}{\mathbb{P}(|C(D)| \geq 1)}.$$

Candidate-level error rates are probabilities under Q, which gives each die weight 1, so they are not die-level rates; only Result 4 is a die-level statement. The positive rate under Q is $\pi = 0.257$ (2.71 candidates per die), so the trivial classifier errs $\min(\pi, 1-\pi) = 0.257$; a bound on the 0-1 error says something only below that.

Pools and regimes. All data come from separately seeded pools, and every choice is made on the validation pool. Two regimes differ only in defect contrast. In the *separable* one the best threshold on peak contrast makes 0 errors on 200,000 candidates; in the *overlapping* one, the main setting, it errs 0.1069 (Figure 1). All results use $\delta = 0.05$.

pool	used for	size used
feature-fit / validation	PCA, scaling / every choice (B, grid)	20,000 / 10,000 candidates
train	a fresh block per repeat	up to 5,200,000 candidates
population / test	$L(h^\star)$, $\theta^\star$ / measured errors, H, G	1,000,000 / 100,000 candidates
certification / far-test	Result 4, defect-free dies	1,600,000 / 100,000 dies

3 Result 1: a threshold on peak contrast

$\mathcal{H} = \{h_\theta\}$ with $h_\theta(x) = +1$ iff $x > \theta$, on the scalar peak contrast x (the maximum of the difference image over the candidate). The learner is exact ERM: sort the n training values, sweep the $n+1$ cuts with cumulative sums, and put $\hat{\theta}$ at the midpoint of the chosen gap. $L(h) = \mathbb{P}_Q(h(x) \neq y)$ and $\hat{L}_S(h)$ is the training error.

$\mathcal{H}$ has VC dimension $d = 1$: one point is shattered (θ below or above it), but for $x_1 < x_2$ the labelling $(+1, -1)$ would need $x_1 > \theta \geq x_2$. By the Sauer–Shelah lemma the growth function, the number of labellings $\mathcal{H}$ induces on $m \geq d$ points, satisfies $\Pi_{\mathcal{H}}(m) \leq (em/d)^d$; for thresholds, $\Pi_{\mathcal{H}}(2n) \leq 2en$.

Theorem 3.1 (Result 1). *Let $\hat{h}$ be any ERM hypothesis on n i.i.d. examples from Q, and $h^\star \in \arg\min_{\mathcal{H}} L$.*

(i) (Realizable.) If $L(h^\star) = 0$ and $0 < \delta \leq 1/2$, then with probability at least $1 - \delta$

$$L(\hat{h}) \leq \frac{2}{n \ln 2}\Big(\ln(2en) + \ln\frac{2}{\delta}\Big).$$

(ii) (Agnostic.) With probability at least $1 - \delta$

$$L(\hat{h}) - L(h^\star) \leq \sqrt{\frac{32\big(\ln(2en) + \ln(4/\delta)\big)}{n}}.$$

Both parts compare the training sample S with an independent ghost sample S' of size n.

Lemma 3.2 (Uniform convergence)**.** *For any class $\mathcal{H}$ of binary classifiers and $n\epsilon^2 \geq 2$,*

$$\mathbb{P}\Big(\sup_{h\in\mathcal{H}} \Big|L(h) - \hat{L}_S(h)\Big| > \epsilon\Big) \leq 4\,\Pi_{\mathcal{H}}(2n)\, e^{-n\epsilon^2/8}.$$

Proof. The argument goes back to Vapnik and Chervonenkis [5]; see also Shalev-Shwartz and Ben-David [3]. *Symmetrization.* On the event, pick (as a function of S) an h with $\Big|L(h) - \hat{L}_S(h)\Big| > \epsilon$. Given S, $\hat{L}_{S'}(h)$ is a mean of n independent indicators with mean $L(h)$ and variance at most $1/(4n)$, so by Chebyshev $\mathbb{P}(\Big|\hat{L}_{S'}(h) - L(h)\Big| \geq \epsilon/2) \leq 1/(n\epsilon^2) \leq 1/2$. With probability at least $1/2$ the two deviations therefore leave $\Big|\hat{L}_S(h) - \hat{L}_{S'}(h)\Big| > \epsilon/2$, and $\mathbb{P}(\sup_h \Big|L - \hat{L}_S\Big| > \epsilon) \leq 2\,\mathbb{P}(\sup_h \Big|\hat{L}_S - \hat{L}_{S'}\Big| > \epsilon/2)$. *Swaps.* All $2n$ points are i.i.d., so swapping the i-th points of S and S' does not change the joint law of (S, S'), and after conditioning on the $2n$ points a hypothesis matters only through its labelling of them. Let $\sigma_i = \pm 1$ be independent fair signs that swap pair i when -1. For a fixed labelling with losses $a_i, b_i \in \{0, 1\}$ on the i-th points of S, S', $\hat{L}_S - \hat{L}_{S'} = \frac{1}{n}\sum_i \sigma_i(a_i - b_i)$ is a sum of independent terms, each in an interval of length at most $2/n$, so Hoeffding's inequality gives $\mathbb{P}(|\cdot| > \epsilon/2) \leq 2\exp\big(-2(\epsilon/2)^2/(n(2/n)^2)\big) = 2e^{-n\epsilon^2/8}$. A union bound over the at most $\Pi_{\mathcal{H}}(2n)$ labellings and the factor 2 above give the claim. □

Proof of Theorem 3.1. (i) Let ϵ be the right-hand side of (i) and $A = \{\exists h : \hat{L}_S(h) = 0,\ L(h) > \epsilon\}$. Since $L(h^\star) = 0$, $h^\star$ is consistent with S almost surely, so every ERM hypothesis is consistent and $L(\hat{h}) > \epsilon$ only on A. *Symmetrization.* On A, pick a consistent h with $L(h) > \epsilon$. Given S, its number of mistakes on S' is binomial with mean $nL(h) > n\epsilon$ and variance below $nL(h)$, so by Chebyshev it is below $nL(h)/2$ with probability at most $4/(nL(h)) < 4/(n\epsilon) \leq 1/2$, using $n\epsilon = \frac{2}{\ln 2}(\ln(2en) + \ln\frac{2}{\delta}) \geq \frac{2}{\ln 2}(\ln(2e) + \ln 4) > 8$ for $n \geq 1$, $\delta \leq \frac{1}{2}$. Hence $\mathbb{P}(A) \leq 2\,\mathbb{P}(\exists h : \hat{L}_S(h) = 0,\ h \text{ errs on} \geq n\epsilon/2 \text{ points of } S')$. *Swaps.* Fix the $2n$ points and a labelling with $k \geq n\epsilon/2$ mistakes. If some pair i holds two mistakes, S always contains one. Otherwise the mistakes sit in k different pairs and all of them land in S' with probability $2^{-k} \leq 2^{-n\epsilon/2}$. *Union bound.* $\mathbb{P}(A) \leq 2\,\Pi_{\mathcal{H}}(2n)\, 2^{-n\epsilon/2} \leq 2(2en)\exp\big(-\frac{n\epsilon\ln 2}{2}\big) = 2(2en)\exp\big(-\ln(2en) - \ln\frac{2}{\delta}\big) = \delta$.

(ii) $L(\hat{h}) - L(h^\star) = [L(\hat{h}) - \hat{L}_S(\hat{h})] + [\hat{L}_S(\hat{h}) - \hat{L}_S(h^\star)] + [\hat{L}_S(h^\star) - L(h^\star)]$, and the middle term is ≤ 0 for ERM, so the excess is at most $2\sup_h \Big|L(h) - \hat{L}_S(h)\Big|$. Lemma 3.2 at $\epsilon/2$ with $d = 1$ gives $\mathbb{P}(L(\hat{h}) - L(h^\star) > \epsilon) \leq 4(2en)e^{-n\epsilon^2/32}$, which equals δ at the right-hand side of (ii). The lemma's condition $n(\epsilon/2)^2 = 8(\ln(2en) + \ln\frac{4}{\delta}) \geq 2$ holds. □

In short

1. Thresholds have VC dimension 1, so $\Pi(2n) \leq 2en$.
2. Realizable: a bad consistent h makes $\geq n\epsilon/2$ mistakes on a ghost sample; random swaps put them all there with probability $\leq 2^{-n\epsilon/2}$; union over $2en$ labellings; solve for ϵ: $\tilde{O}(1/n)$.
3. Agnostic: excess $\leq 2\times$ uniform deviation; ghost sample, swaps and Hoeffding give $e^{-n\epsilon^2/32}$; union; solve: $\tilde{O}(1/\sqrt{n})$.
4. Both are worst cases over distributions: some distribution forces each rate, but a given one can be learned faster.

Measured vs. theory (E1). 100 repeats on fresh dies, $n = 2^4, \ldots, 2^{14}$ (nested prefixes). $\hat{h}$ and $h^\star$ (ERM on the population pool) are evaluated on the same test pool, so its noise largely cancels in the excess. Slopes are fitted where the mean is at least $5\times$ the pool noise floor, and their ranges were fixed before running.

regime	fitted slope (points)	range fixed in advance	bound above mean at every n
separable (test error)	-1.72 (3)	$[-1.15, -0.85]$: no	yes
overlapping (excess error)	-0.94 (6)	$[-1.15, -0.45]$: yes	yes

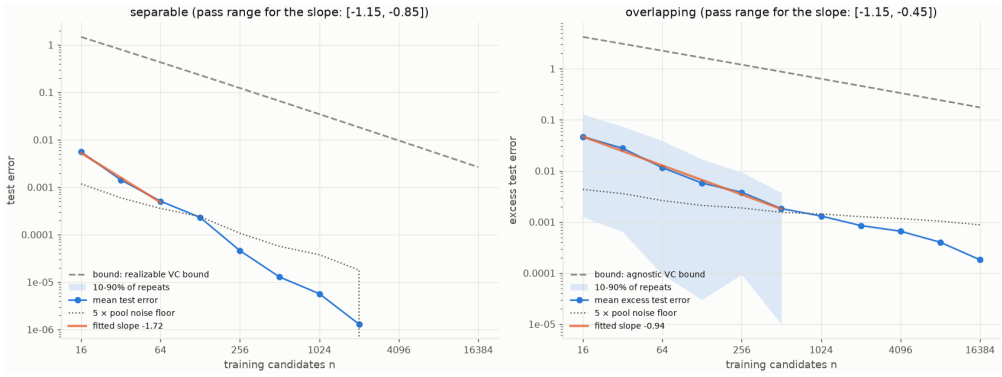


Figure 2: E1: threshold ERM learning curves with the Result 1 bounds (dashed), the fitted slope and 5× the pool noise floor. At $n = 16{,}384$ the agnostic bound is 964× the measured mean excess.

Both bounds hold in every run at every n, which is the test of the theory; the slopes are descriptive. The overlapping excess falls faster than the worst case $n^{-1/2}$: the nuisance density ends sharply at $\theta^\star$, where $P(\text{defect} \mid \text{contrast})$ jumps from 0.27 to 0.88. Such a near-jump at the optimal threshold is consistent with the faster measured rate; this is an observation, not something proved here.

The separable slope check fails. The separable pool has a gap: no candidate has peak contrast in $(0.170, 0.201)$. ERM puts $\hat{\theta}$ midway between the largest nuisance and the smallest defect in its sample, and once that midpoint is in the gap the test error is exactly zero: in 11.0% of repeats at $n = 16$, 81.0% at $n = 1024$, 100.0% at $n = 4096$. The mean is carried by a shrinking set of unlucky runs and falls faster than $1/n$ (fitted on only 3 points). The bound is not contradicted: $1/n$ is forced only by distributions with mass at the boundary, and a margin makes learning faster. I kept the regime as designed and report the failure. The regime is also only nearly realizable: $\theta^\star$ makes 0 errors on $N = 100{,}000$ independent test candidates, so $L(h^\star) \le \ln(1/\delta)/N = 0.00003$ with confidence $1 - \delta$ (as in Section 6), and 1 of the 1,100 training samples is not separable. The proof of (i) only needs $\hat{h}$ to be consistent with its sample, so (i) applies to every other run.

4 Result 2: VC and Rademacher bounds for the linear filter

Feature vectors $x \in \mathbb{R}^{k+1}$ (k features and the intercept coordinate), $\|x\| \le C = 1$; classifiers $x \mapsto \operatorname{sign}(w \cdot x)$; $L_{01}(w) = \mathbb{P}_Q(y\, w \cdot x \le 0)$. The logistic loss of the margin $u = y\, w \cdot x$ is $\ell(u) = \ln(1 + e^{-u})$, and $\psi(u) = \min(\ell(u), \ln 2)$ is its clipped version; $\hat{L}$ denotes training averages. Training always minimizes the logistic loss; ψ is used only to evaluate the bound.

Theorem 4.1 (Result 2)**.** *Let S be n i.i.d. examples from Q.*

(i) (VC.) If $n \ge k + 1$, with probability at least $1 - \delta$, for every $w \in \mathbb{R}^{k+1}$,

$$L_{01}(w) \le \hat{L}_{01}(w) + \sqrt{\frac{8\big((k+1)\ln\frac{2en}{k+1} + \ln\frac{4}{\delta}\big)}{n}}.$$

(ii) (Rademacher, clipped loss.) With probability at least $1 - \delta$, for every w with $\|w\| \le B$,

$$L_{01}(w) \le \frac{1}{\ln 2}\Big(\hat{L}_\psi(w) + \frac{2B}{n}\sqrt{\textstyle\sum_i \|x_i\|^2} + 3\ln 2\sqrt{\frac{\ln(2/\delta)}{2n}}\Big).$$

Applying (ii) at δ/G to each of G grid values of B makes all G hold at once (union bound), so B may be chosen after seeing the data.

Proof. (i) Linear classifiers with an intercept on $\mathbb{R}^k$ have VC dimension $k + 1$: the origin and the k unit vectors are shattered, and by Radon's theorem any $k+2$ points split into two sets whose convex hulls intersect, so they cannot be separated. Our classifiers on $x = (z, 1)$ are exactly this class on $z \in \mathbb{R}^k$ (the convention at $w \cdot x = 0$ does not change this). Lemma 3.2 with Sauer–Shelah, $\Pi(2n) \le (2en/d)^d$ for $d = k + 1 \le n$,

gives $\mathbb{P}(\sup_w \left|L_{01}(w) - \hat{L}_{01}(w)\right| > \epsilon) \le 4(2en/d)^d e^{-n\epsilon^2/8}$. Setting this to δ gives (i); the lemma's condition $n\epsilon^2 \ge 8\ln(4/\delta) \ge 2$ holds.

(ii) Let $\mathcal{F} = \{(x,y) \mapsto \psi(y\, w \cdot x) : \|w\| \le B\}$ and let $\hat{R}_S(\mathcal{F}) = \mathbb{E}_\sigma \sup_{f\in\mathcal{F}} \frac{1}{n}\sum_i \sigma_i f(z_i)$ be its empirical Rademacher complexity, with independent fair signs σ_i.

(a) *Generalization bound.* If every $f \in \mathcal{F}$ takes values in an interval of length Δ, then with probability at least $1-\delta$, for all $f \in \mathcal{F}$, $\mathbb{E}f \le \hat{\mathbb{E}}_S f + 2\hat{R}_S(\mathcal{F}) + 3\Delta\sqrt{\ln(2/\delta)/(2n)}$. Proof: changing one point changes $\Phi(S) = \sup_f(\mathbb{E}f - \hat{\mathbb{E}}_S f)$ by at most Δ/n, so McDiarmid's inequality [2] gives $\Phi(S) \le \mathbb{E}\Phi + \Delta\sqrt{\ln(2/\delta)/(2n)}$ with probability $1-\delta/2$. Symmetrization with a ghost sample and random signs (as in Lemma 3.2) gives $\mathbb{E}\Phi \le 2\mathbb{E}\hat{R}_S(\mathcal{F})$. $\hat{R}_S(\mathcal{F})$ also changes by at most Δ/n per point, so McDiarmid again gives $\mathbb{E}\hat{R}_S \le \hat{R}_S + \Delta\sqrt{\ln(2/\delta)/(2n)}$ with probability $1-\delta/2$. A union bound over the two events gives the claim, which holds for all f at once because it bounds the supremum.

(b) *Range.* ψ takes values in $(0, \ln 2]$, so $\Delta \le \ln 2$ for every B.

(c) *Contraction.* $|\ell'(u)| = 1/(1+e^u) \le 1$, so ℓ is 1-Lipschitz, and so is ψ ($|\min(a,c) - \min(b,c)| \le |a-b|$). By the contraction lemma of Shalev-Shwartz and Ben-David [3, ch. 26] (if $\varphi_1, \dots, \varphi_n$ are ρ-Lipschitz, applying φ_i to the i-th coordinate of every vector in a set $A \subset \mathbb{R}^n$ multiplies its Rademacher complexity by at most ρ; this version has no absolute values and needs no condition $\varphi_i(0) = 0$), applied to the margin vectors $(y_i\, w \cdot x_i)_{i\le n}$ with $\varphi_i = \psi$, $\hat{R}_S(\mathcal{F}) \le \hat{R}_S(\{(x,y) \mapsto y\, w \cdot x\})$.

(d) *Margin class.* For fixed labels, $\sigma_i y_i$ are again independent fair signs, so the margin class has the same complexity as $\{x \mapsto w \cdot x : \|w\| \le B\}$.

(e) *Linear class.* By Cauchy–Schwarz, then Jensen, then $\mathbb{E}[\sigma_i\sigma_j] = \mathbb{1}[i=j]$,

$$\mathbb{E}_\sigma \sup_{\|w\|\le B} \frac{1}{n}\sum_i \sigma_i\, w \cdot x_i = \frac{B}{n}\mathbb{E}_\sigma\Big\|\sum_i \sigma_i x_i\Big\| \le \frac{B}{n}\Big(\mathbb{E}_\sigma\Big\|\sum_i \sigma_i x_i\Big\|^2\Big)^{1/2} = \frac{B}{n}\sqrt{\textstyle\sum_i \|x_i\|^2}.$$

(f) *From ψ to 0-1.* If $u \le 0$ then $\ell(u) \ge \ell(0) = \ln 2$, so $\psi(u) = \ln 2$ and $\mathbb{1}[u \le 0] \le \psi(u)/\ln 2$; hence $L_{01}(w) \le \mathbb{E}\psi/\ln 2$.

Combining (a)–(f) gives (ii). For the grid, each bound fails with probability at most δ/G, so all hold together with probability at least $1-\delta$, in particular for the chosen B. □

Remark 4.2 (Unclipped version). With ℓ in place of ψ only step (b) changes: since $|y\, w \cdot x| \le BC$ and $\ln(1+e^a) - \ln(1+e^{-a}) = a$, ℓ ranges over an interval of length exactly BC, and the last term becomes $3BC\sqrt{\ln(2/\delta)/(2n)}$, which grows with B. It is computed for comparison.

In short

1. VC: half-spaces with intercept in $\mathbb{R}^k$ have $d = k+1$; plug into the uniform-convergence bound $4(2en/d)^d e^{-n\epsilon^2/8} = \delta$. The gap is $\approx \sqrt{d\ln n/n}$: needs $n \gg d$.
2. Rademacher: McDiarmid + symmetrization; clipped logistic loss has range $\le \ln 2$ whatever B; contraction (1-Lipschitz); random signs absorb the labels; Cauchy–Schwarz + Jensen give $\frac{B}{n}\sqrt{\sum \|x_i\|^2}$; a mistake costs $\psi = \ln 2$, so divide by $\ln 2$. No dimension anywhere, only B and the feature norms.
3. A grid of G values of B costs $\ln G$ inside the square root.

Measured vs. theory (E2a). Overlapping regime; 50 repeats on fresh dies, $n = 128, \dots, 32{,}768$. The projected-gradient fit runs at every $B \in \{2^{-1}, \dots, 2^8\}$ ($G = 10$) and B is chosen by validation error; both bounds are evaluated for that filter. The check fixed before running (both bounds above the test error in at least 95% of runs) holds for every feature set.

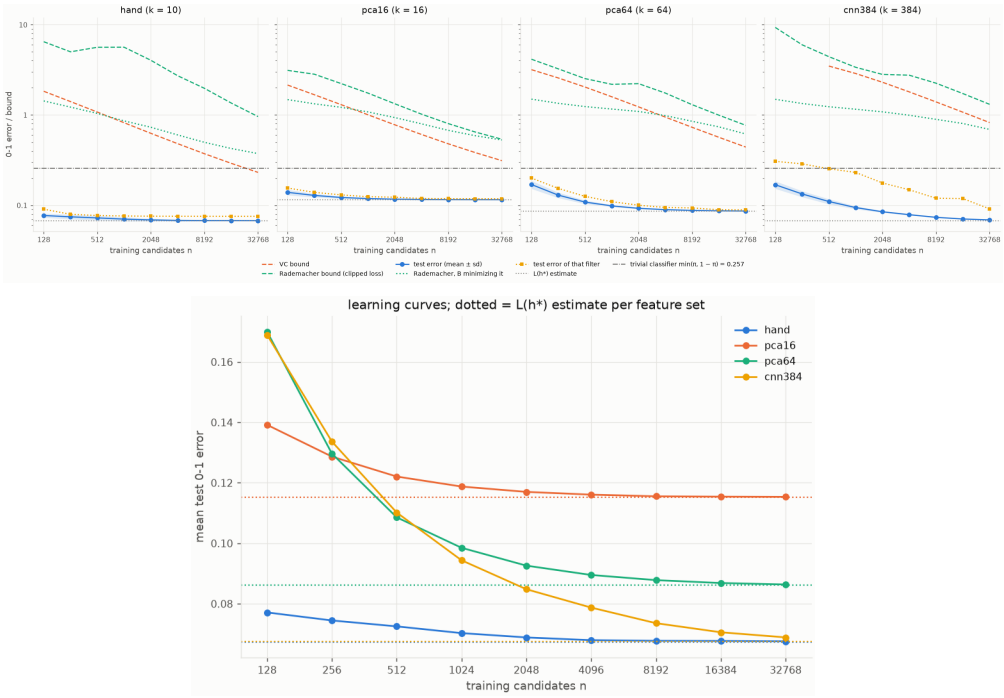


Figure 3: Top (E2a): test error and both Result 2 bounds per feature set; dash-dotted: the trivial error $\min(\pi, 1-\pi)$. Bottom (E2b): learning curves with the $L(h^\star)$ estimates (dotted).

			bound $\geq$ test error		first n with bound < 0.257		at $n = 32{,}768$	
features	k	$L(h^\star)$	VC	Rad.	VC	Rad.	test error	VC / Rad. bound
hand-crafted	10	0.067	100.0%	100.0%	32,768	none in range	0.068	0.23 / 0.96
PCA-16	16	0.115	100.0%	100.0%	none in range	none in range	0.115	0.31 / 0.54
PCA-64	64	0.086	100.0%	100.0%	none in range	none in range	0.086	0.44 / 0.77
CNN-384	384	0.067	100.0%	100.0%	none in range	none in range	0.069	0.83 / 1.31

The VC bound beats the trivial error 0.257 only for the hand-crafted features (from $n = 32{,}768$); for CNN-384 it would need $n \approx 800{,}000$. The usual expectation that the dimension-free bound is tighter fails: at the validated B the Rademacher bound is 0.54–1.31, above VC for every feature set. Choosing B to minimize the Rademacher bound instead trades error for bound; for CNN-384 that reaches 0.69, below VC (0.83), at test error 0.091 instead of 0.069 (Section 7).

Approximation vs. estimation (E2b). $L(h^\star)$ is the test error of the logistic fit on the population pool, an upper estimate of the best-in-class error. Richer CNN features lower it and raise the estimation error at small n (at $n = 128$: 0.024, 0.084, 0.102 for PCA-16, PCA-64, CNN-384), so their curves cross. The ten hand-crafted features match CNN-384's $L(h^\star)$ (0.067 vs. 0.067) with an estimation error of only 0.010 at $n = 128$, and are best at every n tested.

5 Result 3: the learning curve of logistic regression

Unregularized logistic regression: $\hat{\theta}_n = \arg\min_\theta \hat{L}_n(\theta)$, $\hat{L}_n(\theta) = \frac{1}{n}\sum_i \ell(\theta; x_i, y_i)$, $\ell(\theta; x, y) = \ln(1 + e^{-y\,\theta\cdot x})$, $\theta \in \mathbb{R}^p$ ($p = k+1$), ϕ the sigmoid, $L(\theta) = \mathbb{E}_Q \ell(\theta; x, y)$ and $\theta^\star = \arg\min L$. Unlike Results 1 and 2 this is not a bound but a prediction of the learning curve itself.

Theorem 5.1 (Result 3). *Let* $H = \mathbb{E}[\phi(\theta^\star \cdot x)\phi(-\theta^\star \cdot x)\, xx^\top]$ *and* $G = \mathbb{E}[\nabla\ell\, \nabla\ell^\top]$ *at* $\theta^\star$, *with* $\nabla\ell = -\phi(-y\,\theta^\star \cdot x)\, y\, x$. *If* $\theta^\star$ *exists,* $\hat{\theta}_n \to \theta^\star$ *in probability and* $H \succ 0$, *then*

$$\sqrt{n}(\hat{\theta}_n - \theta^\star) \xrightarrow{d} N(0, H^{-1}GH^{-1}), \qquad n\big(L(\hat{\theta}_n) - L(\theta^\star)\big) \xrightarrow{d} \tfrac{1}{2} Z^\top H Z, \quad Z \sim N(0, H^{-1}GH^{-1}),$$

and the limit has mean $\mathrm{Tr}(H^{-1}G)/2$*, which gives the prediction* $\mathbb{E}[L(\hat\theta_n) - L(\theta^\star)] \approx \mathrm{Tr}(H^{-1}G)/(2n)$*. If the model is well specified,* $G = H$ *and the prediction is* $p/(2n)$*, so* $\mathrm{Tr}(H^{-1}G)/p$ *measures misspecification.*

Proof. Derivatives. With $t = \theta \cdot x$, $\frac{d}{dt}\ln(1+e^{-yt}) = -y\,\phi(-yt)$, so $\nabla\ell = -\phi(-y\,\theta\cdot x)\,y\,x$; differentiating again and using $y^2 = 1$ and $\phi(-yt)\phi(yt) = \phi(t)\phi(-t)$, $\nabla^2\ell = g(\theta\cdot x)\,xx^\top$ with $g(t) = \phi(t)\phi(-t)$. Its expectation at $\theta^\star$ is H.

Assumptions. $\theta^\star$ exists because the population is not linearly separable (the classes overlap); training samples that *are* separable have no minimizer and are excluded. Consistency of $\hat\theta_n$ is the standard consistency of M-estimators [4]; here $\hat L_n$ is convex and $\theta^\star$ is unique because $\nabla^2 L \succ 0$ everywhere. $H \succ 0$ because the features are not supported on a hyperplane and $g > 0$. G is finite because $\|\nabla\ell\| \le \|x\| \le 1$. The Hessian is Lipschitz: $g' = g(1-2\phi)$ and $|g'| \le 1/(6\sqrt{3}) \le 1$, so $\left\|\nabla^2\ell(\theta) - \nabla^2\ell(\theta')\right\| \le |(\theta-\theta')\cdot x|\,\|x\|^2 \le \|\theta - \theta'\|$.

Asymptotic normality. Since $\nabla\hat L_n(\hat\theta_n) = 0$, Taylor's theorem in integral form gives $0 = \nabla\hat L_n(\theta^\star) + \bar H_n(\hat\theta_n - \theta^\star)$ with $\bar H_n = \int_0^1 \nabla^2\hat L_n(\theta^\star + s(\hat\theta_n - \theta^\star))\,ds$. By the Lipschitz bound, $\left\|\bar H_n - \nabla^2\hat L_n(\theta^\star)\right\| \le \left\|\hat\theta_n - \theta^\star\right\| \to 0$, and $\nabla^2\hat L_n(\theta^\star) \to H$ by the law of large numbers, so $\bar H_n \to H$ in probability and is invertible with probability tending to 1. The summands of $\nabla\hat L_n(\theta^\star)$ are i.i.d. with mean $\nabla L(\theta^\star) = 0$ and covariance G, so by the central limit theorem $\sqrt{n}\,\nabla\hat L_n(\theta^\star) \to N(0, G)$, and by Slutsky's lemma $\sqrt{n}(\hat\theta_n - \theta^\star) = -\bar H_n^{-1}\sqrt{n}\,\nabla\hat L_n(\theta^\star) \to N(0, H^{-1}GH^{-1})$: the sandwich covariance.

Excess risk. $\nabla L(\theta^\star) = 0$ and $\nabla^2 L$ is Lipschitz with constant 1, so with $v = \hat\theta_n - \theta^\star$,

$$\left|L(\hat\theta_n) - L(\theta^\star) - \tfrac{1}{2}v^\top H v\right| \le \tfrac{1}{6}\,\|v\|^3 .$$

Multiplying by n, the remainder is $O_P(n^{-1/2})$ and $n\cdot\frac{1}{2}v^\top H v \to \frac{1}{2}Z^\top HZ$ by the continuous mapping theorem. Its mean is $\frac{1}{2}\mathrm{Tr}(H\,H^{-1}GH^{-1}) = \frac{1}{2}\mathrm{Tr}(H^{-1}G)$. Reading this limit mean as the mean of $n(L(\hat\theta_n) - L(\theta^\star))$ at finite n is the approximation that the experiment tests.

Well specified. If $P(y = 1 \mid x) = \phi(\theta^\star\cdot x)$, then with $t = \theta^\star\cdot x$, $\mathbb{E}[\phi(-yt)^2 \mid x] = \phi(t)\phi(-t)^2 + \phi(-t)\phi(t)^2 = \phi(t)\phi(-t)$, so $G = H$ and $\mathrm{Tr}(H^{-1}G) = p$. □

In short

1. Gradient $-\phi(-y\theta\cdot x)yx$, Hessian $\phi\phi\,xx^\top$, Lipschitz because $\|x\| \le 1$.
2. One Newton step from $\theta^\star$: $\hat\theta - \theta^\star \approx -H^{-1}\nabla\hat L(\theta^\star)$; CLT on the gradient (mean 0, covariance G/n) gives the sandwich $H^{-1}GH^{-1}/n$.
3. The excess risk is half the squared H-norm: $\frac{1}{2}\mathbb{E}\,v^\top Hv = \mathrm{Tr}(H^{-1}G)/2n$.
4. Well specified $\Rightarrow G = H \Rightarrow p/2n$.

Measured vs. theory (E3). Overlapping regime; PCA-k CNN features, $k \in \{4, 16, 64\}$; 200 repeats on fresh dies, $n \in \{10, 20, 50, 100, 300\} \times p$. $\theta^\star$ is fitted on 1,000,000 population candidates (which biases the measured excess low by at most 1.9%), H and G on the disjoint test pool. The measured quantity is $n(L_{\text{test}}(\hat\theta_n) - L_{\text{test}}(\theta^\star))$. The check fixed before running, measured/predicted in $[0.9, 1.1]$ at the largest n, holds for all three feature sets.

		prediction for n·excess		measured at $n = 300p$			
features	p	$\mathrm{Tr}(H^{-1}G)/2$	$p/2$	mean ± se	ratio	in ±10% from	Wald cov.
PCA-4	5	2.44	2.5	2.51 ± 0.12	1.026	$50\,p$	0.951
PCA-16	17	9.27	8.5	9.63 ± 0.23	1.039	$300\,p$	0.944
PCA-64	65	34.57	32.5	35.32 ± 0.50	1.022	$100\,p$	0.945

At $n = 10p$ the measured excess is 2.190–4.140 times the prediction, and the ratio enters the ±10% band from $n = 50\,p$ to $300\,p$. At $n = 300p$ the sandwich covariance holds as well (n times the covariance of $\hat\theta_n$ over the repeats has 1.006–1.037 times the trace of $H^{-1}GH^{-1}$), and Wald intervals from each run's own sandwich estimate reach nominal coverage (at $n = 10p$ they under-cover, down to 0.852). The model is mildly misspecified ($\mathrm{Tr}(H^{-1}G)/p = 0.978$–$1.091$), so the shortcut $p/2n$ would be off by under 10%.

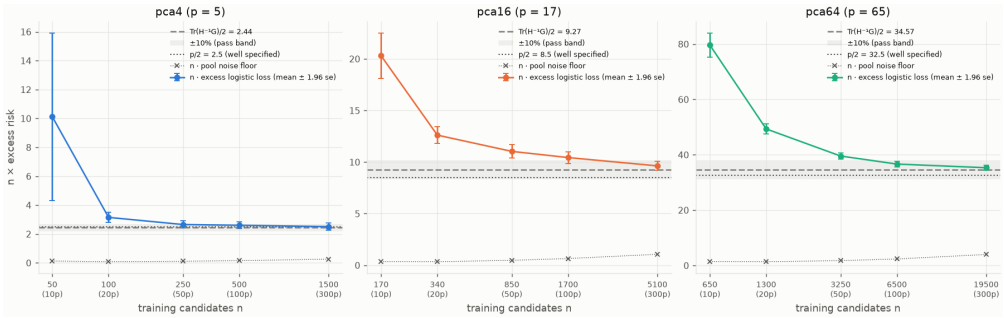


Figure 4: E3: n·excess logistic loss (mean ±1.96 se) with the prediction $\mathrm{Tr}(H^{-1}G)/2$ (dashed), its ±10% band, $p/2$ (dotted) and n times the pool noise floor.

5.1 Result 3b: what more labels will buy

Result 3 needs H and G at $\theta^\star$, which are population quantities. With n labels in hand, the practical question is how much m times as many would help. The difference of two applications of Theorem 5.1 answers it, and the curvature it needs can be estimated from the n labels already collected.

Proposition 5.2 (Result 3b)**.** *Under the assumptions of Theorem 5.1, let $m > 1$ and let the n-sample be the first n points of the mn-sample. Then $\mathbb{E}[L(\hat\theta_n) - L(\hat\theta_{mn})] \approx \frac{\mathrm{Tr}(H^{-1}G)}{2}\left(\frac{1}{n} - \frac{1}{mn}\right)$. With*

$$\hat H_n = \frac{1}{n}\sum_{i\le n} g(\hat\theta_n \cdot x_i)\, x_i x_i^\top, \qquad \hat G_n = \frac{1}{n}\sum_{i\le n} \nabla\ell(\hat\theta_n; x_i, y_i)\nabla\ell(\hat\theta_n; x_i, y_i)^\top,$$

$\mathrm{Tr}(\hat H_n^{-1}\hat G_n) \to \mathrm{Tr}(H^{-1}G)$ in probability, so the plug-in prediction $\hat P_n = \frac{1}{2}\mathrm{Tr}(\hat H_n^{-1}\hat G_n)(\frac{1}{n} - \frac{1}{mn})$ satisfies $n\hat P_n \to \frac{m-1}{2m}\mathrm{Tr}(H^{-1}G)$. It predicts the expected improvement, not that of a single run.

Proof. Difference of two excess risks. $L(\hat\theta_n) - L(\hat\theta_{mn}) = [L(\hat\theta_n) - L(\theta^\star)] - [L(\hat\theta_{mn}) - L(\theta^\star)]$; take expectations and apply Theorem 5.1 at n and at mn. This is linearity of expectation, so the nesting does not affect the mean; it only makes the two terms positively correlated, which lowers the variance of the measured difference.

Consistency of the plug-in. Write $\hat H_n(\theta)$ for the average at θ. Since $|g'| \le 1$ and $\left\|x_i x_i^\top\right\| = \|x_i\|^2 \le 1$, $\left\|\hat H_n(\theta) - \hat H_n(\theta')\right\| \le \|\theta - \theta'\|$, hence $\left\|\hat H_n(\hat\theta_n) - H\right\| \le \left\|\hat\theta_n - \theta^\star\right\| + \left\|\hat H_n(\theta^\star) - H\right\|$. The first term tends to 0 by consistency, the second by the law of large numbers (bounded i.i.d. summands with mean H). For $\hat G_n$, $\nabla\ell\nabla\ell^\top = \phi(-y\,\theta\cdot x)^2 xx^\top$ and $\left|\frac{d}{dt}\phi(-t)^2\right| = 2\phi(-t)^2\phi(t) \le \frac{8}{27} \le 1$ (the maximum of $2a^2(1-a)$, at $a = \frac{2}{3}$), so the same two steps give $\hat G_n \to G$. Since $H \succ 0$, $(A, B) \mapsto \mathrm{Tr}(A^{-1}B)$ is continuous at (H, G), and the continuous mapping theorem gives the claim.

Expected, not single-run. By Theorem 5.1, n times one run's excess at n converges to the non-degenerate $\frac{1}{2}Z^\top HZ$, and likewise at mn. Part of the improvement's spread comes from the $(m-1)n$ labels not yet drawn, which are independent of the first n, so no function of the first n labels can predict it. □

In short

1. Excess at n minus excess at mn: $\frac{\mathrm{Tr}(H^{-1}G)}{2}(\frac{1}{n} - \frac{1}{mn})$, i.e. $\frac{3}{8}\mathrm{Tr}(H^{-1}G)/n$ for $m = 4$.
2. Plug in $\hat H_n, \hat G_n$ at $\hat\theta_n$: Lipschitz in θ plus a law of large numbers at $\theta^\star$ makes them consistent.
3. It predicts the mean; one run keeps an $O(1/n)$ spread that the first n labels cannot remove.

Measured vs. theory (E3b). Same setup, 200 repeats, $n \in \{10, 20, 50, 100\} \times p$, $m = 4$. $\hat H_n, \hat G_n$ use the n labels only; the measured improvement is $L_{\mathrm{test}}(\hat\theta_n) - L_{\mathrm{test}}(\hat\theta_{4n})$. The check fixed before running, mean prediction over mean measurement in $[0.80, 1.25]$ at the largest n, holds for all three feature sets.

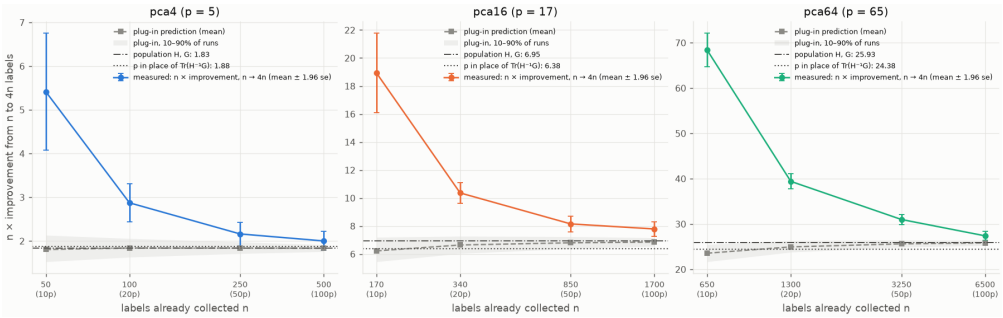


Figure 5: E3b: $n \cdot [L_{\text{test}}(\hat{\theta}_n) - L_{\text{test}}(\hat{\theta}_{4n})]$ (mean ± 1.96 se), the plug-in prediction (mean and 10–90% of runs), the same prediction with the population H, G (dash-dotted) and with p in place of $\text{Tr}(H^{-1}G)$ (dotted).

	n·improvement at $n = 100\,p$				per-run error
features	measured	plug-in	population H, G	plug-in / measured	sd / mean
PCA-4	2.00 ± 0.11	1.84	1.83	0.92	0.78
PCA-16	7.79 ± 0.27	6.85	6.95	0.88	0.49
PCA-64	27.39 ± 0.53	25.83	25.93	0.94	0.27

Estimating H and G from the n labels costs almost nothing: plug-in and population predictions agree within 0.04 (in ratio) at every n. What fails at small n is the asymptotic formula itself: at $n = 10\,p$ the prediction is only 0.33–0.34 of the improvement, because the excess at n is still far above its limit. Single runs scatter: the per-run error is as wide as the spread of the improvement itself, while the prediction varies by only 0.02–0.03 of the mean. For PCA-4, 5.0% of runs even got worse with 4× the labels.

6 Result 4: certifying the die-level false-alarm rate

A frozen stage-2 scorer gives every candidate a score, and a die is flagged at threshold t iff some candidate's score exceeds t. $p(t)$ is the probability that a defect-free die is flagged, and $\hat{p}_m(t)$ the fraction of m i.i.d. defect-free certification dies that are flagged.

Theorem 6.1 (Result 4)**.** *For a grid $t_1 < \cdots < t_T$ fixed before the certification dies are drawn, with probability at least $1 - \delta$, simultaneously for all j,*

$$p(t_j) \le \hat{p}_m(t_j) + \sqrt{\frac{\ln(T/\delta)}{2m}}.$$

So the selected threshold $\hat{t}$, the lowest t_j whose right-hand side is at most α, satisfies $p(\hat{t}) \le \alpha$ with probability at least $1 - \delta$ (if no t_j qualifies, nothing is certified).

Proof. Fix j. The flags $X_i = \mathbb{1}[\text{die } i \text{ flagged at } t_j]$ are independent Bernoulli$(p(t_j))$, because the dies are i.i.d. and the scorer and grid are fixed. Hoeffding's inequality [1] for the $[0,1]$-valued variables $1 - X_i$ gives $\mathbb{P}\big(\frac{1}{m}\sum_i (1 - X_i) - (1 - p(t_j)) \ge \epsilon\big) \le e^{-2m\epsilon^2}$, that is, $\mathbb{P}(p(t_j) - \hat{p}_m(t_j) \ge \epsilon) \le e^{-2m\epsilon^2}$. By the union bound over the T grid points, $\mathbb{P}(\exists j : p(t_j) - \hat{p}_m(t_j) \ge \epsilon) \le T e^{-2m\epsilon^2}$, which equals δ at $\epsilon = \sqrt{\ln(T/\delta)/(2m)}$. On the complementary event every t_j obeys the inequality, in particular the data-dependent $\hat{t}$, whose right-hand side is $\le \alpha$ by construction. □

In short

1. Each threshold's false-alarm count is a sum of i.i.d. coin flips: one-sided Hoeffding, $e^{-2m\epsilon^2}$.
2. The threshold is chosen from the data, so protect all T at once: union bound, $T e^{-2m\epsilon^2} = \delta$. Choosing costs $\ln T$ under the square root.
3. The grid must be fixed before the certification dies are seen (here: from the validation pool).

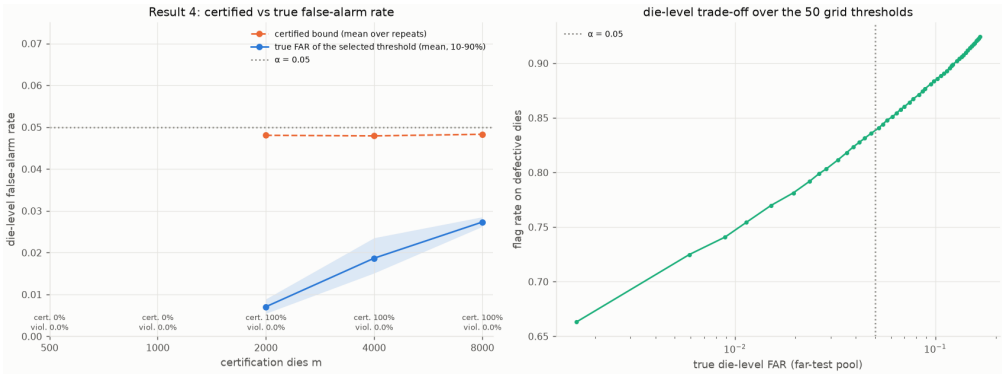


Figure 6: E4: certified bound and true false-alarm rate of the selected threshold vs. m (left; annotations: fraction of repeats that certified and that violated), and the die-level trade-off over the grid (right).

Measured vs. theory (E4). Overlapping regime; $\alpha = 0.05$, $\delta = 0.05$, $T = 50$. Fixed before any certification die is drawn: the scorer (norm-constrained logistic regression on the hand-crafted features, trained once on 4,096 candidates, $B = 128$ by validation, validation error 0.070) and the grid (quantiles of the validation pool's nuisance scores). Each of $R = 200$ repeats uses fresh defect-free dies. The true false-alarm rate is the flag frequency on the far-test pool (100,000 dies). The check fixed before running, a violation fraction $\leq \delta$ at every m, holds.

m	margin $\sqrt{\ln(T/\delta)/2m}$	certifiable	certified bound	true FAR	defective dies flagged
500	0.0831	0.0%	–	–	–
1,000	0.0588	0.0%	–	–	–
2,000	0.0416	100.0%	0.0480	0.0071	0.727
4,000	0.0294	100.0%	0.0479	0.0187	0.779
8,000	0.0208	100.0%	0.0483	0.0273	0.801

Certified bound, true FAR and detection are means over the repeats that certified.

No repeat selected a threshold whose true rate exceeds α (largest: 0.0325). Given the scorer, grid and far-test pool, the repeats' violation indicators are i.i.d. Bernoulli(q), and zero violations have probability $(1-q)^R \leq e^{-qR} < \delta$ whenever $q > \ln(1/\delta)/R$; so with confidence $1-\delta$ the violation rate is at most $\ln(1/\delta)/R = 1.5\%$, below δ as the theorem says.

Nothing can be certified below $m = 1{,}382$, where the margin alone exceeds α; $m = 2{,}000$ certifies in every repeat. Since the rule leaves room for the margin, the selected thresholds land far below α, and more dies buy sensitivity: detection 0.727 at $m = 2{,}000$ and 0.801 at $m = 8{,}000$, against 0.836 for the threshold one would pick knowing $p(t)$.

7 Where the bounds are loose

Every guarantee held in every experiment, and the asymptotic prediction matched. The bounds are still loose by very different factors, and each gap traces back to one step of its proof.

Result 1, agnostic: the worst case. At $n = 16{,}384$ the bound is 964× the measured mean excess. (i) *Rate:* it decays like $\sqrt{\ln n/n}$ because it must hold for every distribution, while the measured excess decays like $n^{-0.94}$. (ii) *Constants:* the factor 2 for the excess and the ghost-sample halving give the 32 in $e^{-n\epsilon^2/32}$. (iii) *Counting:* the union bound runs over all $2en$ labellings of the double sample, though only thresholds near $\theta^\star$ ever compete.

Result 2, VC: dimension against sample size. The gap grows like $\sqrt{(k+1)\ln n/n}$, so the bound says nothing until n is many times $k+1$. For the ten hand-crafted features it is 3.4× the measured error at $n = 32{,}768$, the tightest bound here; for CNN-384 it stays above the trivial error over the whole range.

Result 2, Rademacher: the norm of the weights. At $n = 32{,}768$ and the validated B (mean 128 for the hand-crafted features, $\|w^\star\| = 119$), the hand-crafted bound 0.96 splits into training loss 0.16, complexity term 0.77 and deviation term 0.029 (CNN-384: 0.17, 1.12, 0.029; unclipped bounds 6.35 and 7.22). Clipping keeps the McDiarmid term at 0.029 whatever B; with the logistic loss it pays the range BC and alone reaches 1.32–5.83. The binding term is now the complexity term, which is large because B is: a filter with near-optimal logistic loss needs $\|w\|$ near $\|w^\star\|$, which is large because the classes overlap, and validation picks B of that order. The surrogate costs too: even with infinite data the bound for $w^\star$ cannot go below $L_\psi(w^\star)/\ln 2 = 0.16$–$0.27$, against $L(h^\star) = 0.067$–0.115. Up to constants the VC term is $\sqrt{(k+1)\ln n/n}$ and the Rademacher term $Br/\sqrt{n}$, so Rademacher wins only when Br is small next to $\sqrt{(k+1)\ln n}$. Here Br is 48.4 vs. 10.7 (hand-crafted), 15.0 vs. 13.3 (PCA-16), 33.5 vs. 26.0 (PCA-64) and 70.1 vs. 63.3 (CNN-384). The dimension-free bound suits high-dimensional features with large-margin data, not this overlapping problem.

Result 3: not a bound. The prediction is a limit, so it cannot be "loose". It is within 10% from $n = 50\,p$ to $300\,p$ depending on the features, and underestimates the excess at $n = 10p$ by up to $4.140\times$. Result 3b shows that estimating H and G is not the weak point; the onset of the asymptotic regime is.

Result 4: a worst-case Bernoulli and a union bound. (i) Hoeffding's inequality holds for every Bernoulli parameter, so its margin is the one for variance $\frac{1}{4}$, while flags with p below $\alpha = 0.05$ have variance below α. I kept to elementary tools (Hoeffding rather than variance-adaptive or exact binomial intervals); either would give a tighter margin. (ii) The union over $T = 50$ thresholds widens the margin by $\sqrt{\ln(T/\delta)/\ln(1/\delta)} = 1.52$. The cost shows up as sensitivity: detection 0.727 at $m = 2{,}000$ and 0.801 at $m = 8{,}000$, against 0.836 with full knowledge of $p(t)$.

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